

Paper 5.50 - Oloid Path Carrier Claim Contract

CQE-paper-05.50

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-05.50.md

Paper 5.50 - Oloid Path Carrier Claim Contract

Purpose

Paper 5.50 defines what counts as a valid claim about the Paper 5 rolling carrier. It protects the verified carrier theorem from being confused with unproven prediction or physical-geometry claims.

Admitted Paper 5 Claims

The following claims are admitted from Paper 5:

```
valid binary input produces a continuous rolling trace
every adjacent pair is one legal roll step
head/tail dyads remain binary
Paper 4 payloads can be carried without changing the path rule
invalid bits are rejected
discontinuous jumps are rejected
Rule 30 prediction is out of scope for this receipt
```

Claim Requirements

Any later paper using the Paper 5 carrier must state:

```
the initial carrier state
the input stream or local step being used
the trace state receiving any payload
whether the payload changes the path rule
which receipt proves legal continuity
whether prediction, geometry, or physics claims are separately proved
```

Linked Receipt

The minimum receipt link for Paper 5 is:

```
paper: CQE-paper-05
theorem: Rolling path carrier
receipt: production/formal-papers/CQE-paper-05/oloid_path_carrier_receipt.json
status: pass
```

The receipt is sufficient for finite path continuity. It is not sufficient by itself for Rule 30 prediction, physical Oloid geometry, Wilson loops, or mass-residue claims.

Boundary Failures

The following are boundary failures:

```
accepting a nonbinary input bit
accepting a skipped rolling successor as continuous
allowing payload to mutate roll(q,b)
using a carried payload as proof of the payload's downstream claim
presenting prediction as closed by this carrier receipt
```

Boundary failures are rejected or routed to obligations.

Hidden-Guess Contract

When hidden-guess diagnostics are enabled, the answer must be recorded before the receipt is revealed. The guess record should include:

```
prompt  
predicted step/dyad/scope result  
revealed receipt  
match/mismatch  
next obligation if mismatch
```

The "does this prove prediction?" prompt is required because the correct answer is negative.

Conclusion

Paper 5.50 lets later papers import the rolling carrier honestly. It preserves the path-continuity theorem while keeping prediction and physical geometry as separate proof obligations.